Revised Inquiry Proposal

Icosian Omi-Ring Proof Kernel with E8/Fano Compatibility

0. Purpose

This inquiry proposes a formal Coq proof program for the Omi-Ring using an icosian-inspired arithmetic kernel.

The goal is not to import the old geometric-consensus project unchanged.

The goal is to extract the deterministic geometry from it and rebuild it under OMI doctrine.

The old project explored:

```text id="pb9edm" E8 root systems Weyl reflections Fano-plane consensus BQF / QQF forms Hopf fibrations pinch and branch points quaternion algebra APIs E8 theta-series links

````
The OMI proof target should keep:

```text id="trp55m"
finite incidence
quadratic forms
root systems
reflection/canonicalization
quaternionic arithmetic
norm preservation
deterministic relation stepping
````

But it must quarantine or remove:

```text id="bvpqot" hash identity Merkle identity signature-as-identity BIP32-as-identity content-address-as-protocol-authority

````
OMI identity remains the addressed place-value relation.

1. What the old project contributes

The uploaded documents show several reusable ingredients.

1.1 E8 root system
````
````

````
The old E8 lattice module implements:

```text id="e3s0jr"
240 E8 roots
8 simple roots
Weyl generators
Weyl orbit operations
p-adic height features
quadratic-form integration
````

This is useful because OMI needs a formal way to speak about balanced high-dimensional relation movement.

For Coq, the E8 root system gives a finite target:

```coq id="o4ebf4" Definition E8Root := Vector Q 8.

Definition e8_roots : list E8Root := ...

````
Candidate theorem:

```coq id="hil65l"
Theorem e8_roots_count :
  length e8_roots = 240.
````

Candidate theorem:

```coq id="m1nak3" Theorem e8_roots_norm_two : forall r, In r e8_roots -> dot r r = 2.

````
1.2 Weyl reflections

The old E8 manifold document defines Weyl reflection as:

```text id="q1fsja"
s_α(v) = v - 2(v·α)/(α·α) α
````

This is directly formalizable.

Coq target:

```coq id="jpq8r3" Definition reflect (alpha v : Vec8 Q) : Vec8 Q := v - scale ((2 * dot v alpha) / dot alpha alpha) alpha.

`````
Theorem:

```coq id="kj8wnq"
Theorem reflect_preserves_norm :
  forall alpha v,
    root alpha ->
    norm (reflect alpha v) = norm v.
`````

This is useful for OMI because reflection gives deterministic canonicalization without hashing.

1.3 Fano incidence

The geometric-consensus docs define a 7-point Fano structure with 7 points, 7 lines, 3 points per line, and incidence/block-design comparison.

For OMI, this should be rebuilt as finite incidence proof:

```coq id="uxys1l" Definition Point7 := Fin.t 7. Definition Line7 := list Point7.

Definition fano_lines : list Line7 := ...

`````
Theorems:

```coq id="16niy1"
Theorem fano_two_points_one_line :
  forall p q,
    p <> q ->
    exists! l,
      In l fano_lines /
      In p l /
      In q l.
`````

This supports OMI's gauge-family and Karnaugh-map interpretation because it proves finite incidence structure.

1.4 Quaternion / quadratic form API

The old API reference includes quaternion algebra, norm forms, Hilbert symbols, BQF/TQF/QQF operations, and QQF-to-E8 theta linkage.

For OMI, this suggests the icosian bridge should pass through:

```text id="0n7cff" Golden field quaternions over Q($\sqrt{5}$) quaternion norm icosian units quadratic-form compatibility

```
This is better than jumping directly to octonions.

2. OMI correction layer

The old project used hash, content address, BIP32, Merkle root, and signature
binding as structural identity devices.

OMI cannot use those as identity.

So every old component must be classified:

```text id="yjh8f7"
Allowed in OMI core:
  relation
  incidence
  norm
  closure
  reflection
  deterministic traversal
  finite enumeration
  proof of equality by structure

Carrier-only / outside identity:
  hash
  signature
  Merkle root
  BIP32 path
  SHA-derived tetrahedron
  content address
```

Therefore, the Coq proof must not prove:

```text id="sli4fk" hash equality implies identity signature implies identity Merkle root implies accepted state

```
Instead it must prove:

```text id="vwf2xm"
structural resolution implies identity
relation step is deterministic
gauge action preserves norm
finite incidence is decidable
validation predicate is decidable
```
```

3. Revised proof ladder

The Coq development should be staged as follows.

Stage 1 — Finite vectors and exact arithmetic

Define exact rational vectors.

```coq id="ct88yr" Definition Vec (n : nat) := Vector.t Q n.

~~~~
Needed operations:

```coq id="mhscw3"
dot
add
neg
scale
norm
~~~~

Theorems:

```coq id="99fwlp" dot_comm dot_add_l norm_nonnegative_if_ordered_later

~~~~
This gives the base for E8 and reflection.

### Stage 2 — Golden field

Define:

```coq id="9ad8zr"
Record Golden := {
  gr : Q;
  gs : Q
}.
~~~~

Interpreted as:

```text id="6z2mim" gr + gs√5

~~~~
Define:

```coq id="y4oqmh"
g_add
~~~~

````
g_neg
g_mul
g_conj
g_norm
````

Theorems:

```coq id="cvk7oy" g_add_assoc g_mul_assoc g_mul_comm g_distrib g_conj_involutive g_norm_mul

````
Stage 3 — Golden quaternions

Define:

```coq id="jc3qzh"
Record GQuat := {
  qr : Golden;
  qi : Golden;
  qj : Golden;
  qk : Golden
}.
````

Define Hamilton multiplication.

Theorems:

```coq id="vepamv" q_mul_assoc q_distrib_l q_distrib_r q_conj_involutive q_norm_mul

````
Important negative doctrine:

```text id="wm668r"
Do not prove q_mul_comm.
Quaternion multiplication is not commutative.
````

Stage 4 — Unit icosians

Define the 120 unit icosians as a finite list.

Theorems:

```coq id="9xm94y" unit_icosians_length : length unit_icosians = 120.

unit_icosians_norm_one : forall x, In x unit_icosians -> q_norm x = 1.
```

unit_icosians_closed_mul : forall x y, In x unit_icosians -> In y unit_icosians -> In (q_mul x y) unit_icosians.

unit_icosians_inverse : forall x, In x unit_icosians -> In (q_conj x) unit_icosians.

````
This proves the finite local rotation/action group.

### Stage 5 — Icosian span / ring candidate

Define finite integer spans of unit icosians.

```coq id="1x2trj"
Record IcosianSpan := {
  terms : list (Z * GQuat);
  terms_are_units :
    forall z q, In (z,q) terms -> In q unit_icosians
}.
````

Prove closure under addition, negation, and multiplication.

```coq id="7x1n4u" span_closed_add span_closed_neg span_closed_mul

````
This is the ring-candidate layer.

Stage 6 — E8 compatibility

Use the icosian coordinate split:

```text id="5lfojc"
4 quaternion coordinates
×
2 golden-field components
=
8 rational coordinates
````

Define:

```coq id="la5l30" icosian_to_vec8 : GQuat -> Vec 8.

````
Then prove that selected norm behavior corresponds to an E8-like quadratic form.

Initial modest theorem:

```coq id="9hk40v"
````
````

```
Theorem icosian_vec8_norm_compatible :
  forall q,
    vec8_norm (icosian_to_vec8 q) = golden_norm_projection (q_norm q).
```

Later stronger theorem:

```coq id="uh8zmu" Theorem e8_roots_from_icosian_units : forall u, In u selected_icosian_roots -> In (icosian_to_vec8 u) e8_roots.

```
This connects the icosian bridge to the uploaded E8 work.

Stage 7 — Fano / gauge compatibility

Define the OMI gauge family as 16 controls.

```coq id="73h615"
Definition OmiGauge := Fin.t 16.
```

Define a map:

```coq id="mrdk2c" gauge_to_unit : OmiGauge -> GQuat.

```
Then define an OMI-ring step:

```coq id="6wpy7i"
Definition omi_ring_step (s : GQuat) (g : OmiGauge) : GQuat :=
  q_mul s (gauge_to_unit g).
```

Core theorems:

```coq id="arubor" omi_ring_step_deterministic : forall s g x y, x = omi_ring_step s g -> y = omi_ring_step s g -> x = y.

omi_ring_step_preserves_norm : forall s g, q_norm (omi_ring_step s g) = q_norm s.

omi_ring_step_closed : forall s g, IsOmiRingState s -> IsOmiRingState (omi_ring_step s g).

```
Then connect the 16 gauge controls to finite incidence if desired:

```coq id="i1f9y3"
```

```
gauge_fano_projection :
  OmiGauge -> option Point7.
```

This should be optional, not foundational.

Stage 8 — Relational quotation

Define quoted relation:

```coq id="g9464g" Record QuotedRelation := { qr_source : GQuat; qr_gauge : OmiGauge; qr_target : GQuat }.

````
Define adjudication:

```coq id="lms84j"
Definition adjudicates (qr : QuotedRelation) : Prop :=
  qr_target qr = omi_ring_step (qr_source qr) (qr_gauge qr).
````

Theorem:

```coq id="fwf9uj" adjudication_decidable : forall qr, { adjudicates qr } + { ~ adjudicates qr }.

````
This is the formal version of:

```text id="7qcm5u"
The Tetragrammatron can adjudicate the quoted relation.
````

4. What to reuse from the uploaded project

Reuse as proof targets

```text id="x69xr1" E8 root count = 240 Weyl reflection formula Fano incidence structure quaternion norm API BQF / QQF vocabulary theta-series coefficients as reference tests Hopf fibration as projection mental model pinch/branch points as singularity vocabulary

````
Do not reuse as OMI identity

```text id="vfwtyq"
BIP32 path identity
SHA content addressing
hash-to-tetrahedron
Merkle root identity

````

```
signature-bound “who”
cryptographic spine as authority
```

These may remain carrier metadata or external interoperability, but not OMI core identity.

5. Proposed Coq files

```text id="pgzcen" coq/VecQ.v coq/GoldenField.v coq/GoldenQuaternion.v coq/IcosianUnits.v coq/IcosianSpan.v coq/E8Roots.v coq/WeylReflection.v coq/FanoIncidence.v coq/OmiRingIcosian.v coq/OmiRingQuotation.v

````
Dependency order:

```text id="27tm2r"
VecQ
  ↓
GoldenField
  ↓
GoldenQuaternion
  ↓
IcosianUnits
  ↓
IcosianSpan
  ↓
E8Roots / WeylReflection / FanoIncidence
  ↓
OmiRingIcosian
  ↓
OmiRingQuotation
````

6. First realistic Coq milestone

The first milestone should be small:

```text id="93z3um" GoldenField.v GoldenQuaternion.v

````
Prove:

```coq id="wo2k68"
g_mul_assoc
q_mul_assoc
q_norm_mul
````

This gives the arithmetic foundation.

7. Second realistic Coq milestone

Define the 120 unit icosians as a list and prove:

```coq id="307fq8" length unit_icosians = 120 forall u, In u unit_icosians -> q_norm u = 1

````
Closure under multiplication can initially be proven by finite table lookup.

8. Third realistic Coq milestone

Define 16 OMI gauge actions as selected unit icosian actions.

Prove:

```coq id="madnxg"
omi_ring_step_deterministic
omi_ring_step_preserves_norm
adjudication_decidable
````

This is the first OMI-native proof result.

9. Canonical inquiry lock

```text id="fagkvl" The icosian bridge is investigated as a candidate arithmetic proof kernel for the Omi-Ring.

The uploaded E8, Fano, quadratic-form, and geometric-consensus materials are treated as prototype geometry and implementation precedent.

OMI adopts only the deterministic structural parts: finite incidence, exact arithmetic, norm preservation, Weyl reflection, quadratic reduction, and decidable adjudication.

OMI does not adopt hash, signature, BIP32, Merkle, or content-address fields as protocol identity.

Identity remains the addressed place-value relation.

The Coq target is to prove that selected icosian/golden-quaternion gauge actions are closed, deterministic, norm-preserving, and adjudicable as quoted OMI relations. ```
```